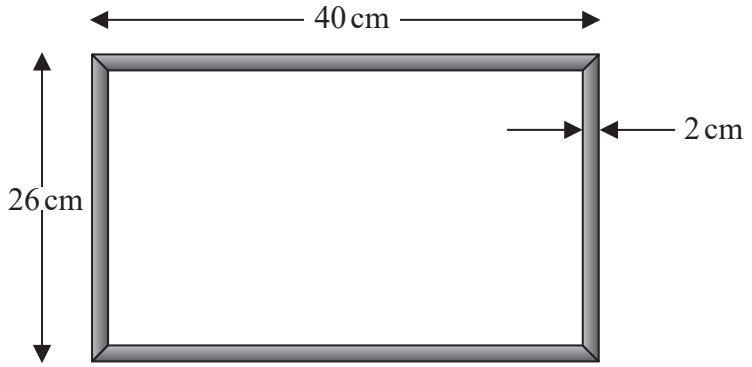
19 Peter has 4 photos.
Each photo is in the shape of a rectangle with dimensions 12 cm by 8 cm.

Peter is going to enlarge each photo by a scale factor of 1.5
He wants to put all 4 enlarged photos in the picture frame shown below.



The dimensions of the outside of the picture frame are 40 cm by 26 cm.
The width of each side of the frame is 2 cm.

Can the 4 enlarged photos be put in the picture frame so they do **not** overlap?
You must show how you get your answer.

1 mark
Enlarge lengths

	L	W
	12	8
×1.5		×1.5
	18	12

Each photo: 18cm × 12cm.

L=length; W=width.

Inside frame: 40 − 4 = 36cm; 26 − 4 = 22cm.

1 mark
Subtract two borders

Inside area = 36 × 22 = 792 cm^2.

1 mark
Compare total areas

Four photo areas = 4 × 18 × 12 = 864 cm^2.

No: 864 > 792, so there is not enough area.

1 mark
Cannot fit

(Total for Question 19 is 4 marks)